**Clinical Study Summary Report (CSSR) Document Status:** Final | **Version:** 1.0 | **Date:** 06-Apr-2026

---

## 1. Administrative & Study Identification

**Full Study Title:** An Investigational Immuno-therapy Study of Temozolomide Plus Radiation Therapy With Nivolumab or Placebo, for Newly Diagnosed Patients With Glioblastoma (GBM, a Malignant Brain Cancer) **Short Title/Acronym:** CheckMate 548 **Protocol Number:** CA209-548 **NCT Number:** NCT02667587 **Posting ID:** 924926836131243437 **Sponsor Name / Company:** Bristol Myers Squibb (BMS) **Investigational Product:** Nivolumab (Opdivo), Temozolomide (Temodar) **Phase of Development:** Phase 3 **Trial Status:** COMPLETED **Medical Monitor:** Bristol Myers Squibb Clinical Operations Lead

---

## 2. Study Synopsis & Rationale

**2.1 Study Objective Primary Objective:** To evaluate progression-free survival (PFS) and overall survival (OS) of adding Nivolumab to the current standard of care (Temozolomide [TMZ] and Radiation Therapy [RT]) versus placebo in patients with newly diagnosed glioblastoma with methylated MGMT promoter. **Secondary Objectives:** To evaluate investigator-assessed PFS and OS rates at specific time points (12 and 24 months).

**2.2 Background & Rationale** Glioblastoma (GBM) is an aggressive primary malignant tumor of the central nervous system with a very poor prognosis and a five-year survival rate of less than five percent. Standard treatment includes surgery followed by radiation and chemotherapy, but treatment options have remained limited. This study aims to evaluate whether the addition of the immune checkpoint inhibitor nivolumab (Opdivo) can improve progression-free and overall survival outcomes in newly diagnosed patients whose tumors are MGMT-methylated.

---

## 3. Study Design & Methodology

**3.1 Design Framework Study Type:** Interventional **Control Type:** Placebo-controlled **Blinding:** Single Blind **Randomization:** 1:1

**3.2 Planned Enrollment & Duration Target $N$:** 716 patients **Number of Sites:** 123 locations globally **Estimated Study Start:** 09-May-2016 **Estimated Primary Completion:** 22-Dec-2020

---

## 4. Subject Selection (Eligibility Criteria)

**4.1 Inclusion Criteria** 1. Males and Females, age $\ge 18$ years old. 2. Newly diagnosed brain cancer or tumor called glioblastoma or GBM. 3. Karnofsky performance status of $\ge 70$ (able to take care of self) and substantial recovery from surgery resection. 4. Tumor test result shows MGMT methylated or indeterminate tumor subtype.

**4.2 Exclusion Criteria** 1. Biopsy-only of GBM with less than 20% of tumor removed. 2. Prior treatment for GBM (other than surgical resection). 3. Any known tumor outside of the brain, or recurrent/secondary GBM. 4. Active known or suspected autoimmune disease.

---

## 5. Intervention & Treatment Plan

**5.1 Investigational Regimen Dosage/Frequency:** Nivolumab (240 mg every 2 weeks $\times$ 8, then 480 mg every 4 weeks) OR Placebo + RT (60 Gy over 6 weeks) + TMZ (Temozolomide, Trade Name: Temodar, ATC Code: L01AX03) (75 $mg/m^2$ once daily during RT, then 150-200 $mg/m^2$ once daily on days 1-5 of every 28-day cycle $\times$ 6). **Route of Administration:** Intravenous (Nivolumab/ Placebo), Oral (Temozolomide), and external beam Radiation Therapy. **Duration of Treatment:** Until disease progression, unacceptable toxicity, or standard trial completion timeframe (evaluated up to ~35 months after randomization).

**5.2 Concomitant & Prohibited Therapy Permitted:** Standard concomitant medications for symptomatic management. **Prohibited:** Use of concurrent investigational therapies and other specific immunosuppressive regimens or immune checkpoint therapies not defined in the protocol.

---

## 6. Study Timeline & Visit Schedule

| Visit ID | Window (Days) | Key Activities/Procedures |
| :--- | :--- | :--- |
| **Screening** | Day -28 to 0 | Informed Consent, Medical History, Surgery, MGMT Methylation Testing, Karnofsky Performance Status |
| **Baseline** | Day 0 | Randomization, First Dose of Nivolumab/Placebo + RT + TMZ |
| **Interim** | Every 2 to 4 weeks | Safety Labs, Efficacy Assessment (MRI, RANO criteria), Nivolumab infusion |
| **End of Study** | Upon Progression/Toxicity | Final Assessment, Exit Interview, Survival Follow-up |

---

## 7. Outcome Measures (Endpoints)

**7.1 Primary Endpoint Definition:** Progression-Free Survival (PFS) and Overall Survival (OS) assessed in the randomized population with no corticosteroids at baseline, and in the overall randomized population. **Measurement Method:** PFS determined by Blinded Independent Central Review (BICR) based on Radiologic Assessment in Neuro-Oncology (RANO) criteria. OS defined as time from randomization to the date of death by any cause.

**7.2 Secondary Outcomes** 1. Overall Survival (OS) Rates at 12 Months. 2. Overall Survival (OS) Rates at 24 Months. 3. Progression Free Survival (PFS) Based on Investigator Assessment.

---

## 8. Safety & Adverse Event Reporting

**Adverse Event (AE) Monitoring:** Standard collection via physical examinations, laboratory assessments, and monitoring of Grade 3/4 treatment-related adverse events. **Serious Adverse Events (SAEs):** Reported within 24 hours of site awareness. **Data Monitoring Committee (DMC):** An independent Data Monitoring Committee conducted routine reviews to evaluate the safety and efficacy endpoints (e.g., advising unblinding when primary OS endpoint was not met).

---

## 9. Statistical Analysis Plan (SAP) Summary

**Analysis Populations:** Intent-to-Treat (ITT) overall randomized population, and a specific sub-population evaluation for patients without baseline corticosteroid usage. **Sample Size Justification:** Powered to detect statistically significant improvements in PFS and OS (Hazard Ratio metrics). **Handling of Missing Data:** Standard time-to-event censoring rules applied; patients who have not died or progressed by the end of the study will be censored to the last known date alive or last tumor assessment date.

---

## 10. Quality Assurance & Regulatory Compliance

**GCP Compliance:** This study was conducted in accordance with Good Clinical Practice (GCP) guidelines, the International Conference on Harmonisation, and the Declaration of Helsinki. **Monitoring Plan:** Periodic tracking of protocol adherence via institutional review boards (IRB) and independent ethics committees at each site. **Audit Trail:** Clinical data captured via electronic case report forms (eCRFs) monitored by the central sponsor system.

---

## 11. Study Identifiers & Governance

**Primary ID:** CA209-548 **Registry Number:** NCT02667587 **Posting ID:** 924926836131243437 **Sponsor/Lead Organization:** Bristol Myers Squibb **Study Title:** CheckMate 548 **Official Regulatory Title:** A Randomized Phase 3 Single Blind Study of Temozolomide Plus Radiation Therapy Combined With Nivolumab or Placebo in Newly Diagnosed Adult Subjects With MGMT-Methylated (Tumor O6-methylguanine DNA Methyltransferase) Glioblastoma

---

## 12. Clinical Framework (Glioblastoma Specific)

**Primary Condition:** Glioblastoma (GBM) **Disease Areas:** Brain Neoplasms **Preferred UMLS Name:** Glioblastoma Multiforme **Diagnosis Threshold:** Newly diagnosed GBM, MGMT-methylated or indeterminate. **Medical Methodology:** Surgical resection followed by standard of care Chemoradiotherapy + Immune Checkpoint Inhibition. **Optimization Target:** Delaying tumor progression (PFS) and improving overall survival (OS).

---

## 13. Study Procedures & Methodology

**Management Pathway:** Standard oncology clinical pathway for GBM. **Education Protocol (HCP):** RECIST and RANO criteria assessment training for radiological reviews. **Education Protocol (Patient):** Informed consent and specific training regarding neuro-oncological treatment schedules. **Quality Audit Mechanism:** Blinded Independent Central Review (BICR) for impartial and accurate progression tracking.

---

## 14. Observation & Follow-Up Schedule

**Total Duration:** Follow-up for up to 35 months or until death/ withdrawal. **Onsite Visit Cadence:** Every 2 to 4 weeks (aligned with Nivolumab and Temozolomide administration cycles). **Remote Monitoring Frequency:** Regular survival tracking via standard reporting protocols. **Checklist Review Interval:** Routine MRI reviews according to RANO criteria (post-radiotherapy windows).

---

## 15. Sign-off & Approval

| Role | Name | Signature | Date | | :--- | :--- | :--- | :--- | | **Principal Investigator** | [Macarena De La Fuente / Relevant PI] | ____ **| [DD-MMM-YYYY] | | Clinical Operations Lead | [Name] | ___ |** ***[DD-MMM-YYYY] | | Lead Statistician | [Name] |*** _____ | [DD-MMM-YYYY] |